PRACTICE REPORT

COMPUTER NETWORKS

Lab 5

CONFIGURING NETWORK DEVICES

Teacher: **Phan Trung Phát**

Group: CS4283.Q12.CTTT

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OTHER EVALUATIONS (*):

Content	Result
Average total time to complete the lab (1)	9 Hours
Video Link (2) *(if any)*	
Feedback (3) *(if any)* + Challenges + Suggestions ...	
Self-Assessment Score (4)	10/10

(*): Sections (1) and (4) are mandatory.

TASK ASSIGNMENT TABLE:

Members	Task Details
Nguyễn Dương Phúc	Task 1, 2, 3 (Work - 100%)

The section below is the detailed report of the group/individual responsible.

WARNING

FOR INTERNAL CIRCULATION ONLY

Question 1

1. Evidence:

Figure 1. Detailed IP Address Configuration

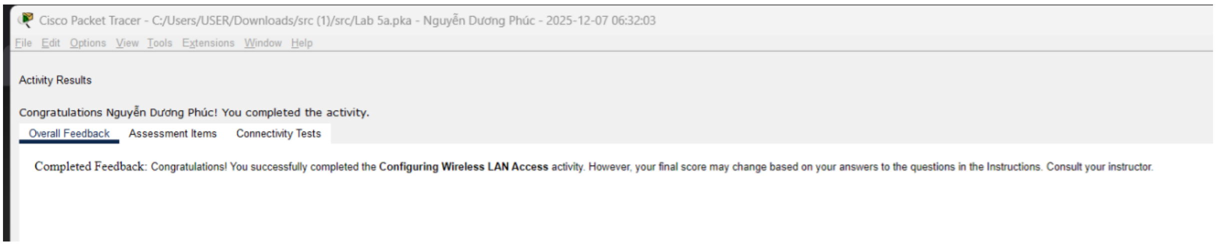

Figure 2. Overall Results Task 1

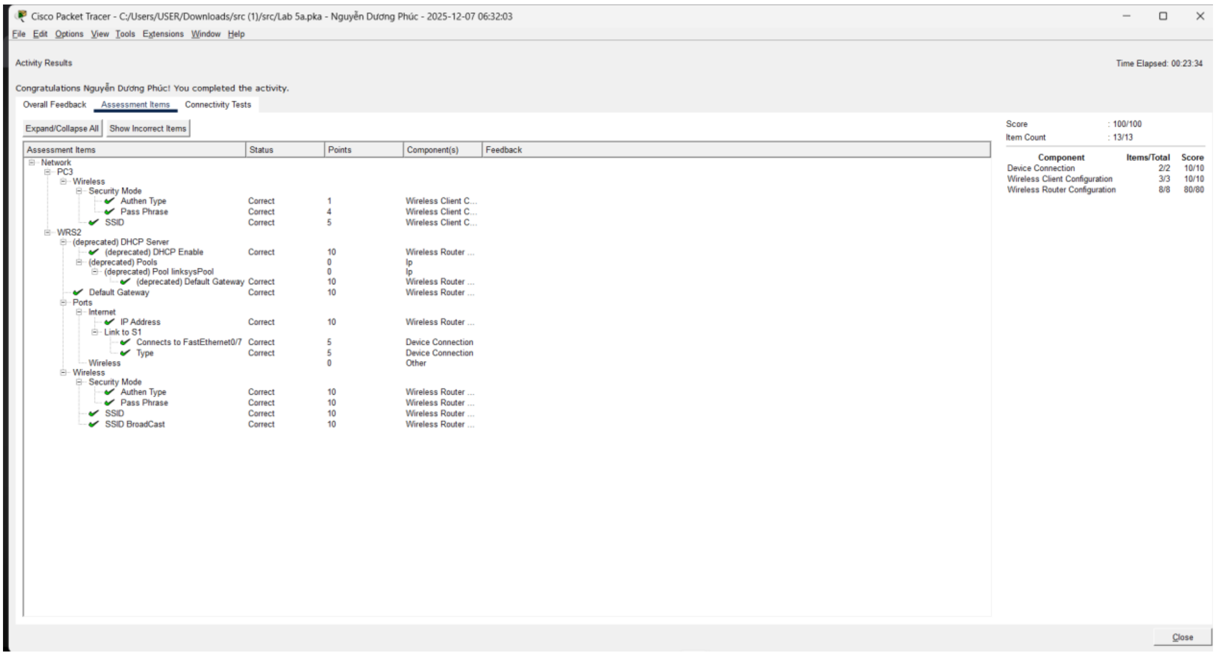

Figure 3. Detailed Results Task 1

Question 2

1. Evidence:

- **The configurations of router R1:**

```
[OK]
R1#show run
Building configuration...

Current configuration : 1318 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname R1
!
!
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO1941/K9 sn FTX15240P9D
!
!
!
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
```

1

```
interface GigabitEthernet0/0
 ip address 192.168.10.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 ip address 192.168.11.1 255.255.255.0
 duplex auto
 speed auto
!
interface Serial0/0/0
 ip address 209.165.200.225 255.255.255.252
 clock rate 64000
!
interface Serial0/0/1
 no ip address
 clock rate 2000000
 shutdown
!
interface FastEthernet0/1/0
 switchport mode access
 switchport nonegotiate
 shutdown
!
interface FastEthernet0/1/1
 switchport mode access
 switchport nonegotiate
 shutdown
!
interface FastEthernet0/1/2
 switchport mode access
 switchport nonegotiate
 shutdown
!
interface FastEthernet0/1/3
 switchport mode access
 switchport nonegotiate
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
router eigrp 1
 network 192.168.10.0
 network 192.168.11.0
 network 209.165.200.0
 auto-summary
!
ip classless
!
ip flow-export version 9
```

2

```
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
 password 7 0822455D0A16
 login
!
line aux 0
!
line vty 0 4
 login
!
!
!
end


R1#
```

3

Figure 4. R1 - Show run Command

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.10.1    YES manual up                    up
GigabitEthernet0/1     192.168.11.1    YES manual up                    up
Serial0/0/0            209.165.200.225 YES manual up                    up
Serial0/0/1            unassigned      YES unset  administratively down down
FastEthernet0/1/0      unassigned      YES unset  administratively down down
FastEthernet0/1/1      unassigned      YES unset  administratively down down
FastEthernet0/1/2      unassigned      YES unset  administratively down down
FastEthernet0/1/3      unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

Figure 5. R1 - Show IP interface brief Command

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter are
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

D    10.0.0.0/8 [90/2170112] via 209.165.200.226, 00:05:31, Serial0/0/0
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0
     192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.11.0/24 is directly connected, GigabitEthernet0/1
L       192.168.11.1/32 is directly connected, GigabitEthernet0/1
     209.165.200.0/24 is variably subnetted, 3 subnets, 3 masks
D       209.165.200.0/24 is a summary, 00:11:03, Null0
C       209.165.200.224/30 is directly connected, Serial0/0/0
L       209.165.200.225/32 is directly connected, Serial0/0/0

R1#
```

Figure 6. R1 - Show IP route Command

- **The configurations of router R2:**

```
[OK]
R2#show run
Building configuration...

Current configuration : 927 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname R2
!
!
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO1941/K9 sn FTX1524E7C4
!
!
!
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
```

1

```
interface GigabitEthernet0/0
 ip address 10.1.1.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 ip address 10.1.2.1 255.255.255.0
 duplex auto
 speed auto
!
interface Serial0/0/0
 ip address 209.165.200.226 255.255.255.252
!
interface Serial0/0/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
router eigrp 1
 network 10.0.0.0
 network 209.165.200.0
 auto-summary
!
ip classless
!
ip flow-export version 9
!
!
!
no cdp run
!
!
!
!
!
line con 0
 password 7 0822455D0A16
 login
!
line aux 0
!
line vty 0 4
 login
!
!
!
end
```

2

Figure 7. R2 - Show run Command

```
R2#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.1.1.1        YES manual up                    up
GigabitEthernet0/1     10.1.2.1        YES manual up                    up
Serial0/0/0            209.165.200.226 YES manual up                    up
Serial0/0/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R2#
```

Figure 8. R2 - Show IP interface brief Command

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
D       10.0.0.0/8 is a summary, 00:07:59, Null0
C       10.1.1.0/24 is directly connected, GigabitEthernet0/0
L       10.1.1.1/32 is directly connected, GigabitEthernet0/0
C       10.1.2.0/24 is directly connected, GigabitEthernet0/1
L       10.1.2.1/32 is directly connected, GigabitEthernet0/1
D    192.168.10.0/24 [90/2170112] via 209.165.200.225, 00:13:31, Serial0/0/0
D    192.168.11.0/24 [90/2170112] via 209.165.200.225, 00:10:42, Serial0/0/0
     209.165.200.0/24 is variably subnetted, 3 subnets, 3 masks
D       209.165.200.0/24 is a summary, 00:07:59, Null0
C       209.165.200.224/30 is directly connected, Serial0/0/0
L       209.165.200.226/32 is directly connected, Serial0/0/0

R2#
```

Figure 9. R2 - Show IP route Command

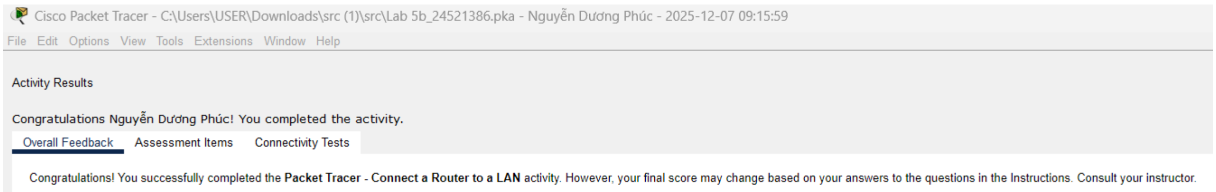

Figure 10. Overall Results Task 2

Figure 11. Detailed Results Task 2

2. Explanation:

- **Show run:**

Displays the entire running configuration of the router.

Used to check IP settings, passwords, interface status, routing protocols, hostname, and other active configurations.

⇨ Shows the device's current configuration.

- **Show IP interface brief:**

Shows a summary of all interfaces, including IP address, status (up/down), and protocol state.

⇨ Used to quickly verify whether interfaces have correct IPs and are operational.

- **Show IP route:**

Displays the router's routing table.

Shows known networks, how routes were learned (C, S, R), next-hop information, and selected paths.

⇨ Used to confirm routing and troubleshoot connectivity issues.

Question 3

1. Evidence:

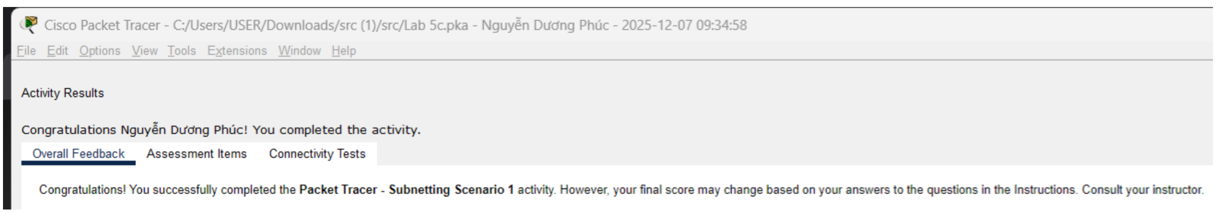

Figure 12. Overall Results Task 3

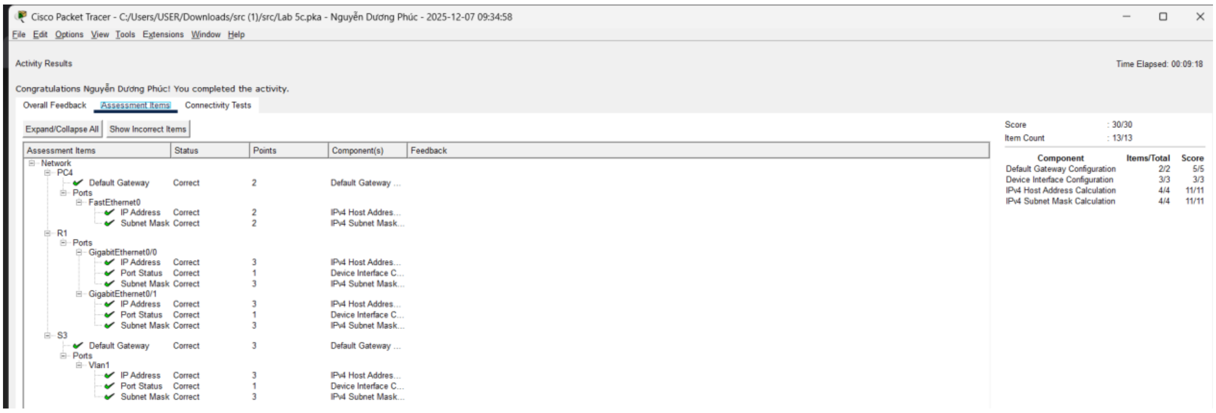

Figure 13. Detailed Results Task 3

2. Explanation:

No.	Network Address	Starting Host Address	Ending Host Address	Broadcast Address
0	192.168.100.0	192.168.100.1	192.168.100.30	192.168.100.31
1	192.168.100.32	192.168.100.33	192.168.100.62	192.168.100.63
2	192.168.100.64	192.168.100.65	192.168.100.94	192.168.100.95
3	192.168.100.96	192.168.100.97	192.168.100.126	192.168.100.127
4	192.168.100.128	192.168.100.129	192.168.100.158	192.168.100.159
5	192.168.100.160	192.168.100.161	192.168.100.190	192.168.100.191
6	192.168.100.192	192.168.100.193	192.168.100.222	192.168.100.223
7	192.168.100.224	192.168.100.225	192.168.100.254	192.168.100.255

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	192.168.100.1	255.255.255.244	N/A
	G0/1	192.168.100.33	255.255.255.244	N/A
	S0/0/0	192.168.100.129	255.255.255.244	N/A
R2	G0/0	192.168.100.65	255.255.255.244	N/A
	G0/1	192.168.100.97	255.255.255.244	N/A
	S0/0/0	192.168.100.158	255.255.255.244	N/A
S1	VLAN 1	192.168.100.2	255.255.255.244	192.168.100.1
S2	VLAN 1	192.168.100.34	255.255.255.244	192.168.100.33
S3	VLAN 1	192.168.100.66	255.255.255.244	192.168.100.65
S4	VLAN 1	192.168.100.98	255.255.255.244	192.168.100.97
PC1	NIC	192.168.100.30	255.255.255.244	192.168.100.1
PC2	NIC	192.168.100.62	255.255.255.244	192.168.100.33
PC3	NIC	192.168.100.94	255.255.255.244	192.168.100.65
PC4	NIC	192.168.100.126	255.255.255.244	192.168.100.97

END